

MME8215: Operations Research - Seminar

V. Decision theory, game theory.

Consider the following problems, for some of them you can find a solution posted on teams.

Problem set

1. You are the trainer of a basketball team playing an inter-university tournament over a weekend. You won the first game of the tournament yesterday, and now you have to decide whether to play with the same starting 5 or make changes in the team. You want to assess your decision considering the possibilities of winning (p_w) or losing (p_l) the game and of having any player injured (p_i) or all healthy (p_h) after the game. Since it is a friendly tournament, your priority is that none of your classmates ends up injured, so you establish the following priorities between the utility values of the possible outcomes: v_1 (victory – no injuries) $> v_2$ (lose – no injuries) $> v_3$ (win – any injured) $> v_4$ (lose – any injured).

- Draw a decision tree indicating the probabilities and utility values of the possible outcomes, considering that the probability of winning or losing and the probability of having any player injured or all healthy are independent and also independent of your decision to make changes in the team.
- How would the decision tree change if you consider that the probability of having any player injured is not independent of your decision to make changes?

2. Ash is a renowned Pokémon trainer. He thinks that his most beloved Pokémon, Pikachu, could have been injured in the last battle of the Winners Trophy play-offs. Therefore, he asks himself: *should he use Pikachu in the next battle against Team Rocket?*

The trainer thinks that there is a probability of 0.1 of victory. However, if Pikachu participates in the fight, his injury could worsen. Ash thinks that there is a probability of 0.2 that Pikachu is actually injured from the last battle.

The utilities (in points) that the trainer employs are:

- if Pikachu fights, and Ash wins the battle, and Pikachu is not injured, +100;
- if Pikachu fights, and Ash wins the battle, and Pikachu is injured, +50;
- if Pikachu fights, and Ash loses the battle, and Pikachu is not injured, 0;
- if Pikachu fights, and Ash loses the battle, and Pikachu is injured, -50;
- If Pikachu does not fight and Pikachu is injured, the utility is -10;

- If Pikachu does not fight and Pikachu is not injured, the utility is 0.

Ash wants to use decision theory to support his decision-making in the battle against Team Rocket.

- a. Draw the decision tree for this problem considering that the probabilities of the result of the battle and the injury are independent.
- b. Determine which is the best decision for Ash.
- c. Ash could acquire further information about the probabilities of the result of the battle. The additional information determines the result of the battle (win or lose), and based on previous results, it has a certain accuracy to successfully predict the result of the battle. Draw again the decision tree to incorporate into the decision problem the acquisition of this imperfect information about the result of the battle, indicating in the corresponding branches the revised probabilities

3. Alice and Bob are two classmates that want to celebrate the end of the course with a game. They will go to their favourite bar and Bob will randomly select a bottle of wine from the wine list. Alice will do a blind testing of the wine, with no possibility to see the wine selected by Bob. If Alice guesses the type of wine (red or white) just by holding her glass (by smell), Bob will pay 20€. Otherwise, if she fails, Alice will pay Bob 10€. She may also taste the wine for 5€ (that is, Bob will discount 5€ of the bet if she guesses the type of wine, or Alice will pay 5€ extra if she does not).

Alice thinks she can use decision theory to improve her chances of winning the bet. She knows that in the wine list of their favourite bar, there are 15 bottles of red wine and 11 of white wine. She has also done some blind wine tasting to prepare for the bet and found out that it is actually not that easy to guess the colour of the wine just by tasting it blindfolded. She guessed correctly $\frac{3}{4}$ of the times she tried a white wine and $\frac{2}{4}$ of the times she tried a red wine.

- a. Draw the decision tree for Alice without taking into account that she may test the wine to get additional information.
- b. Modify the decision tree for Alice now taking into account that she may test the wine.
- c. What is the best alternative for Alice?

4. Your company, *Altered Silicon*, manufactures solar energy panels, and it has received an urgent order for 40 panels from a new German customer which is willing to pay 1000€ per panel. Altered Silicon has committed its production capacity to other orders, and the only alternative left to serve the new order is to subcontract the production involved.

Altered Silicon has located two possible suppliers:

- Supplier A: based in Romania, placed an offer of 500€ per panel
- Supplier B: from Poland, which would sell each panel for 550€

The cost to transport one batch of 40 panels from either of the two suppliers to the German firm is 1000€. Nonetheless, the customer's technical specifications are very demanding, so Altered Silicon estimates that Supplier A could supply one correct batch with a probability of 70%, whereas Supplier B could do it with a probability of 80%. If the batch were incorrect, Altered Silicon could sell it to another customer at the price of 200€ per panel. Altered Silicon is considering the possibility of asking for a test panel from each supplier to test it before placing an order. Altered

Silicon's testing system guarantees that a faulty panel is always detected, while there is a 5% margin of error with correct panels. The testing system is prepared to assess two test panels, and the overall cost (cost of the two panels + transport + cost of the test) comes to 2000€.

- a. Draw the corresponding decision tree and indicate with a letter per branch the associated probability or the associated cost (depending on the case)
- b. Solve the decision tree
- c. Up to how much should Altered Silicon pay for the test?

5. Two manufacturers, M1 and M2, compete for the sales of two different product lines which are equally profitable. In both cases, M2's sales volume triples that of M1. Due to some technological advances, both manufacturers can make substantial improvements to the two products, but they are not sure as to which development and commercialisation strategy they should adopt. If the improvements to the two products are done at the same time, neither manufacturer can have them ready to sell under 12 months. One alternative is to carry out an "intensive programme" to first develop one of the two products and to try to commercialise it before the competition does. If this were the case, M2 can have one product ready to sell in 9 months, while M1 needs 10 months (due to previous commitments in its production installations). Either manufacturer can have the second product ready after another 9-month period. For either product line, if both manufacturers commercialised the improved models simultaneously, it is estimated that M1 can increase the percentage of all the future sales of this product by 8% of the total (from 25% to 33%). Likewise, M1 can increase its total sales in 20, 30 and 40% of the total if it commercialises the product 2, 6 and 8 months before M2 does, respectively. Moreover, M1 can lose 4, 10, 12 and 14% of the total if M2 manages to commercialise it 1, 3, 7 and 10 month(s) before it, respectively.

Formulate this problem as if it were a zero-sum game with two players and then establish the strategy the manufacturers must adopt according to the MinMax criterion.

6. Two students, Alice and Bob, are playing a game named Bluff, that goes as follows:

- Alice writes a number on a piece of paper. Without showing Bob what she has written, she tells him what she wrote. Alice might be lying or telling the truth.
- Then Bob must guess if Alice is lying or telling the truth, or he can choose to pass and not to continue playing.
- If Bob discovers that Alice lied, Alice must pay Bob 50€. If Bob wrongly accuses Alice of lying, Bob pays her 25€. If Alice told the truth and Bob guessed this, then Alice pays Bob 5€. If Alice lied and Bob does not guess that she lied, then Bob pays Alice 24€.
- Should Bob decide to pass, then neither player wins or loses anything.

- a. Consider this a 2-player's zero-sum game and write down the tabular representation.
- b. How would you determine the game mean value and the mixed strategies of both players in this case?
- c. Using a method to calculate the mixed strategies, a solver provides the following solution for Bob ($u_1 = 0, u_2 = 0, u_3 = 0.02$), where u_1 is the decision variable associated to the alternative "Alice lied", u_2 the decision variable associated to the alternative "Alice told the truth" and u_3 the decision variable associated to "Pass". Write down the model and determine his optimal strategy.

7. Suppose there are two political parties, party A and B (with opposite views). They are competing in an election. Each party has three possible strategies: they can focus their campaign on young citizens (below 30), middle-aged citizens (30 – 50), or senior citizens (over 50). The parties use national statistics to estimate the number of votes that can shift in each demographic group. Here are the estimated values for each demographic group:

- Young voters: They represent 60% of the undecided voters.
- Middle-aged voters: They represent 25% of the undecided voters.
- Senior citizens: They represent 15% of the undecided voters.

From past elections, party A has collected the following estimation about undecided voters on each scenario:

- Young voters: If both parties focus on young voters, party A will get 40% of the young voters and party B 60% of the young voters. If party A focuses on the young voters and party B does not, party A will get 70% of the young voters. If party B focuses on the young voters and party A does not, party A will get 10% of the young voters. If neither party focuses on the young voters, party A will get 40% of the young voters.
- Middle-aged voters: If both parties focus on middle-aged voters, party A will get 60% of the middle-aged voters and party B 40% of the middle-aged voters. If party A focuses on the middle-aged voters and party B does not, party A will get 90% of the middle-aged voters. If party B focuses on the middle-aged voters and party A does not, party A will get 30% of the middle-aged voters. If neither party focuses on the middle-aged voters, party A will get 60% of the middle-aged voters.
- Senior citizens: If both parties focus on senior citizens, party A will get 70% of the senior citizens and party B 30% of the senior citizens. If party A focuses on the senior citizens and party B does not, party A will get 90% of the senior citizens. If party B focuses on the senior citizens and party A does not, party A will get 20% of the senior citizens. If neither party focuses on the senior citizens, party A will get 70% of the senior citizens.

Model the strategies of both political parties, following these steps:

1. Identify the alternatives for both parties and draw a tabular representation of the game were Player A is party A.
2. Complete the tabular representation of the game, estimating the total percentage of undecided voters that would vote for each party.
3. Calculate the strategy of each party using the MinMax criteria.